# Measuring the Human, Measuring Society

## Francis Galton and the Anthropometric Sciences

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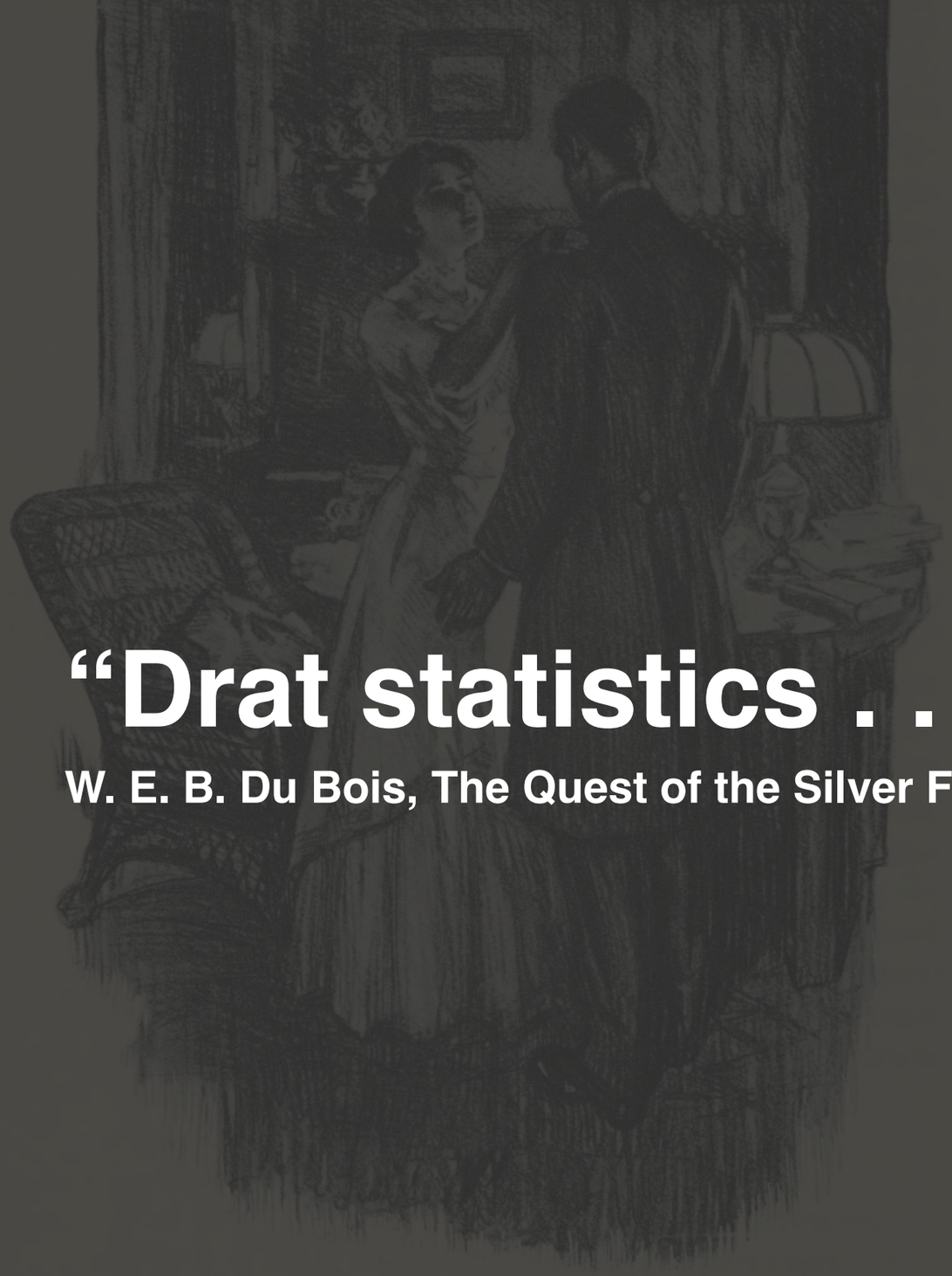
Week 5: October 28, 2026

# Today

– Measuring the Human, Measuring Society
– Mathematics as Infrastructure
– Genealogy of Statistics and Scientific Racism
– Facial Recognition Technology and the Politics of Recognition



SILVER FLEECE

A NOVEL

BY

W. E. BURGHARDT DuBOIS

AUTHOR OF "THE SOULS OF BLACK FOLK"

ILLUSTRATED BY H. S. De LAY

# "Drat statistics . . . these are folks!"

**W. E. B. Du Bois, The Quest of the Silver Fleece (1911)**

"BLES, ALMOST THOU PERSUADEST ME TO BE A FOOL"
[CHAPTER XXIX]

CHICAGO
A. C. McCLURG & CO.
1911

# Today we're going to talk about the fault lines between *statistics* and *folks*

MATH

NATURAL
HISTORY

ART

PHIL-
OSOPHY

MATH

NATURAL
HISTORY

ART

PHIL-
OSOPHY

# Once we have data, what do we do with it?

Our Subject for Today

# How biometric and eugenic projects used measurement and quantification to make racialized claims appear objective and natural

**“Practices of measurement have long been necessitated for the interest of racialism and racialism has always been a form of measurement.” (Dixon-Román, 2025, p. 411)**

# Measuring the Human, Measuring Society

# Quantification and Measurement

### Quantification

The broader process of turning qualities, categories, and experiences into numerical form so they can be counted, compared, ranked, and calculated with. It includes defining what counts, what gets left out, and which categories exist at all.

### Measurement

“the assignment of numerals to objects or events according to rules” (Stevens, 1946). It requires deciding what property is measured, with what instrument, according to what scale, and the statistical operations applicable.

**“Part of the challenge of understanding algorithmic oppression is to understand that mathematical formulations that drive automated decisions are made by human beings.”**

Noble, S. U. (2018). Algorithms of oppression: How search engines reinforce racism. New York University Press. p. 19

# Two Guiding Questions:

- How do you quantify people?
- How do you take the measure of society?

# Mathematics as Infrastructure

# There are several core areas of mathematics that provide machinery for making use of data

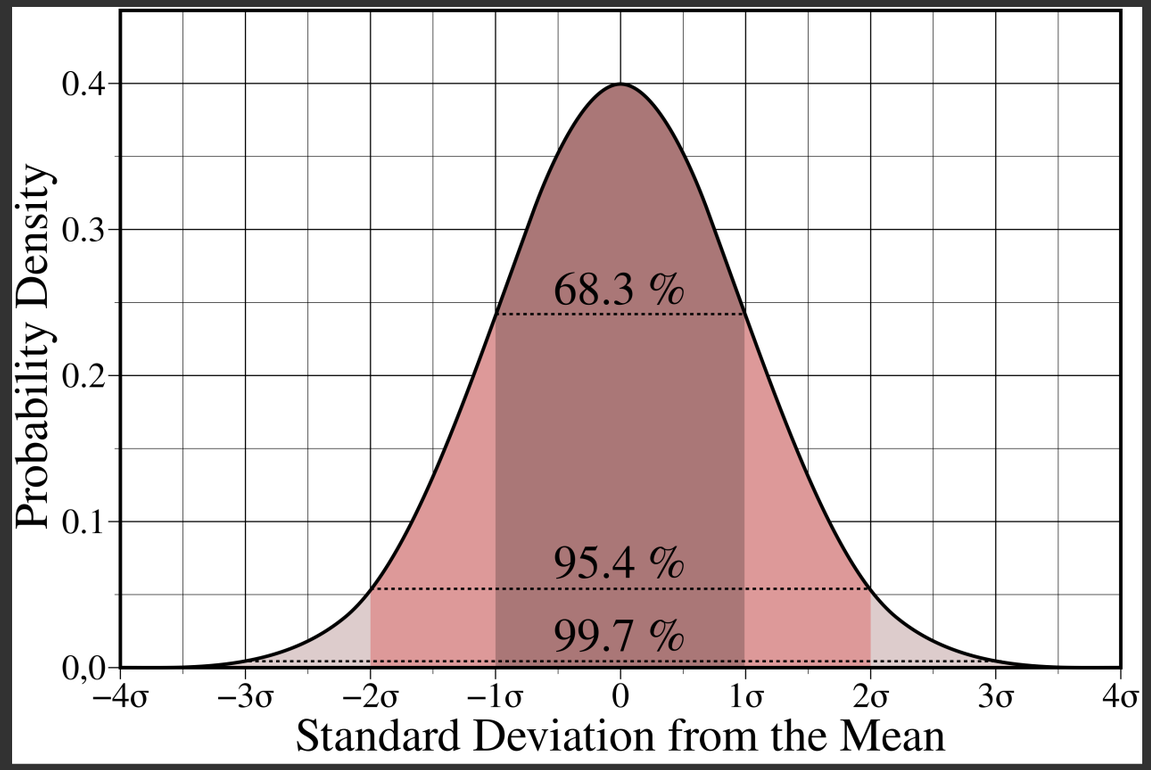
- Statistics
- Probability
- Linear Algebra

# Statistics and Probability

The emergence of statistics and probability was brought about by material, social, and mathematical transformations. They each emerged separately and converged in the 19th century

# Statistics

Statistics is a discipline concerned with collection, analysis, and interpretation of data. Statistics makes use of various methods and practices to conduct analysis and develop inferences based on data.




https://en.wikipedia.org/wiki/Statistics#/media/File:Standard_Normal_Distribution-en.svg

# Probability

Probability is the mathematics, it assigns a number between 0 and 1 to how likely an event is.

$$0 \leq P(A) \leq 1$$

Modern probability began in the 17th century with problems about games of chance and was eventually formalized into mathematical probability theory by Andrey Kolmogorov in 1933.

# Three ways of defining Linear Algebra

| Classification | Definition |
| --- | --- |
| Elementary | Linear Algebra is the study of vectors, matrices, systems of linear equations, and problems that can be solved using linear algebraic techniques |
| Abstract | Axler defines Linear Algebra as "the study of linear maps on finite-dimensional spaces" (Axler, 2015) |
| Computational | "Linear Algebra is the study of vectors and ways of manipulating vectors" (Deisenroth, 2020) |

# There are three core concepts in linear algebra

Scalar $\lambda \in \mathbb{R}$

We can think of a scalar as a single value, usually a real number

Vector

Example Vector $a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3$

Vectors are objects that we add together and multiply by a scalar to get another vector.

Matrix A $m \times n$ matrix is a collection of elements consisting of $m$ rows and $n$ columns. We can think of matrices as a collection of vectors or a linear transformation

Example Matrix

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 8 & 2 & 1 \\ 3 & 3 & 4 \end{bmatrix}$$

# History of Linear Algebra

Linear algebra encompasses many techniques that evolved over time and in several places including China, Europe, Greece, and Egypt

In the 19th century James Joseph Sylvester and Arthur Cayley introduced algebraic techniques for linear algebra and coined the term matrix, which comes from the Latin term for womb

# What does Linear mean?

Linear transformations are functions that map one vector to another

Formal Definition

A transformation $L()$ is called linear if the following properties hold

Addition: $L(v + w) = L(v) + L(w)$

Scalar Multiplication: $L(\lambda v) = \lambda \cdot L(v)$

# What is Algebra?

Algebra comes from the name the text Al-jabr (الجبر) written by polymath Muhammad ibn Musa al-Khwarizmi who worked in Baghdad in the 9th century

Algebra broadly refers to the formal manipulation of abstract symbols

# Linear Algebra and Data

Linear algebra provides tools for representing and working with structured data.

- A vector can represent one observation, with each entry recording a measured feature.
- A matrix can organize many observations and variables in a single structure.
- Matrix operations can combine, compare, and transform data, and help solve systems of equations.
- These tools are used across fields to study relationships and patterns in data, from population tables to images.

How data is organized into vectors and matrices reflects choices about what is measured and how.

# Modes of Reasoning with Data

## Deductive reasoning

Deductive reasoning means moving from premises to conclusions by drawing valid inferences. A valid inference means that it logically follows from the premises

## Inductive reasoning

Inductive reasoning is where generalizations or principles are made from a body of observations. For example, machine learning can be considered a form of automated induction.

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ANTHROPOMETRIC LABORATORY,

*Arranged by Francis Galton, F.R.S.*

| | |
|---|---|
| Sex | Colour of eyes |
| EYESIGHT. | |
| Greatest distance in inches, of reading "Diamond" type | right eye \| left eye |
| Colour-sense, goodness of | |
| JUDGMENT OF EYE | |
| [illegible] divergent line of 15 inches | in three parts \| in five parts |
| [illegible] of estimating squareness | |
| HEARING. | |
| Keenness can hardly be tested here owing to the noises and echoes. | |
| Highest audible note | between 0·000 and 0·000 vibrations per second. |
| BREATHING POWER. | |
| Greatest expiration in cubic inches | |

| | |
|---|---|
| Date | Initials |
| SWIFTNESS | |
| of blow of hand in feet per second | |
| STRENGTH | |
| of squeeze in lbs. of | right hand \| of pull in lbs. |
| | left ,, |
| SPAN OF ARMS | |
| From finger tips of opposite hands | feet, inches. |
| HEIGHT | |
| Sitting, measured from seat of chair | feet, inches. |
| Standing in shoes | feet, inches. |
| less height of heel..... | inches. |
| Height without shoes | feet, inches. |
| WEIGHT | |
| in ordinary in-door clothing in lbs. | |

Age last birthday? ____
Married or unmarried? ____
Birthplace? ____
Occupation? ____
Residence in town, suburb, or country? ____

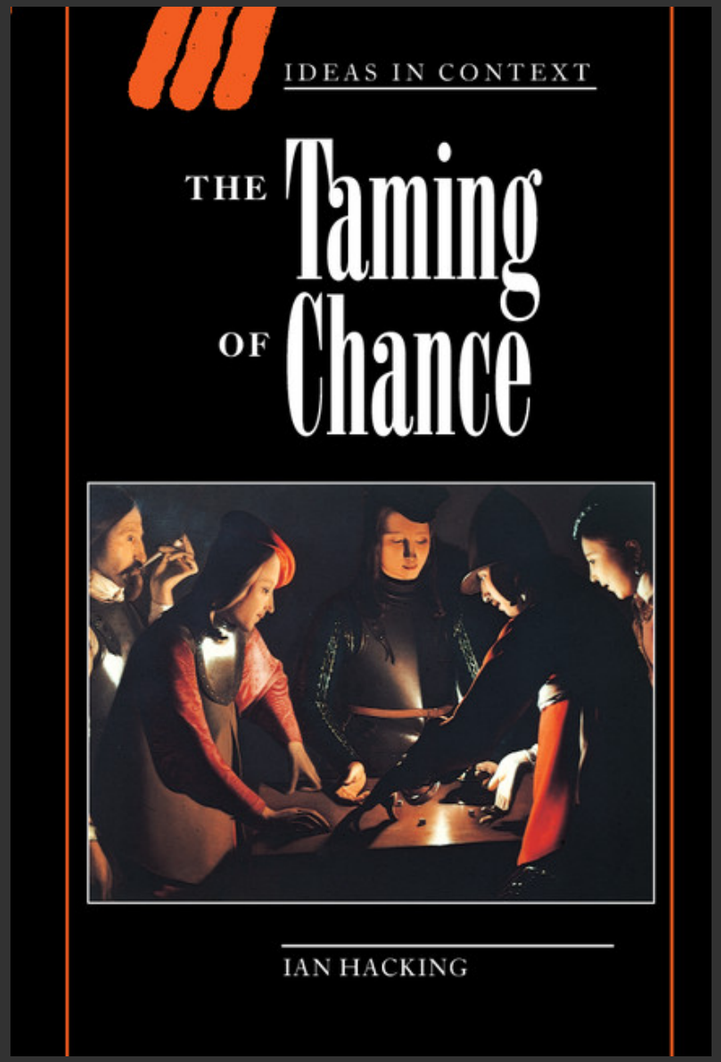
Q 2

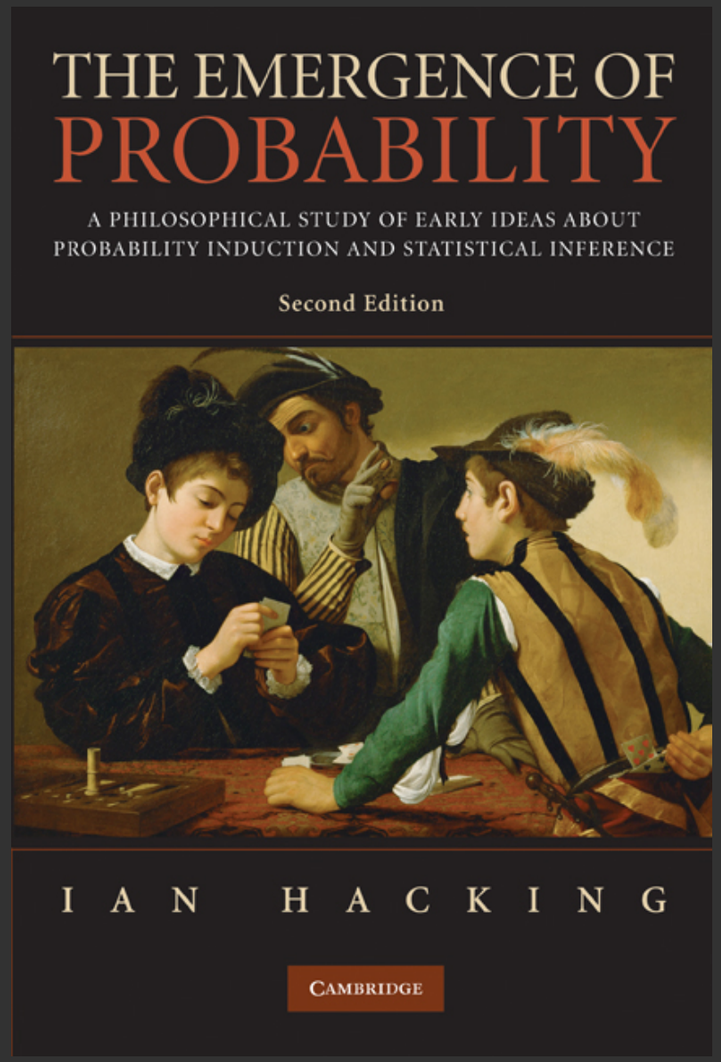
# Genealogy of Statistics and Scientific Racism

## Where Did Statistics Come From?

'Statistik' was coined by Gottfried Achenwall in 1749, and he thought of what he called statistics as the collection of 'remarkable facts about the state'.

Statistics evolved out of the study and quantification of the state: large-scale efforts to conduct censuses, measure populations, and track state-relevant metrics (Hacking, 1990).



“One can tell the story of biopolitics as the transition from the counting of hearths to the counting of bodies. The subversive effect of this transition was to create new categories into which people had to fall, and so to create and to render rigid new conceptualizations of the human being.”

(Hacking, 2015, p. 281)

## Social Statistics and Social Physics

*folks as a collective*

Social Statistics or Social Physics sought to discover immutable statistical laws that governed society

## Biometrics and Anthropometry

*The individual bodies that make up folks*

Emerging from 19th century study of physiognomy and phrenology, the body became a target for quantification and measurement to track norms and deviances

# Key figures

- Adolphe Quetelet
- Francis Galton
- Karl Pearson
- Alphonse Bertillon

# Methods for Quantification and Measurement

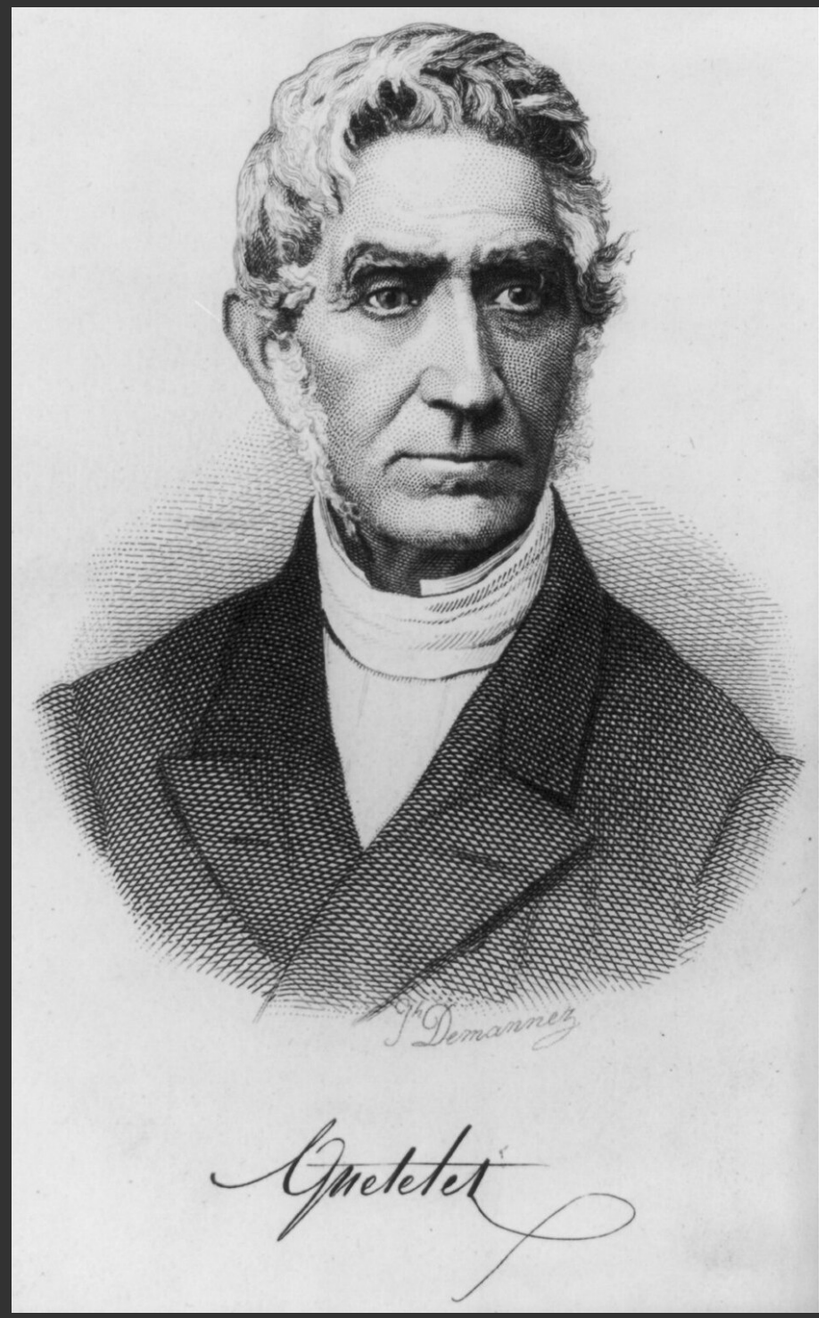
# Adolphe Quetelet (1796-1874)

Belgian statistician, astronomer, and sociologist

Founded the science of anthropometry and created the body mass index (BMI)

developed the notion of the *homme moyen* or average man and was an important thinker in the development of eugenics

Quetelet's move was to treat human traits using probability as if they were measurement errors scattered around a true value, the *homme moyen* or "average man."



# What is an Average

Mathematically we can represent an average as:

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

but an average can also be a knowledge claim based on data

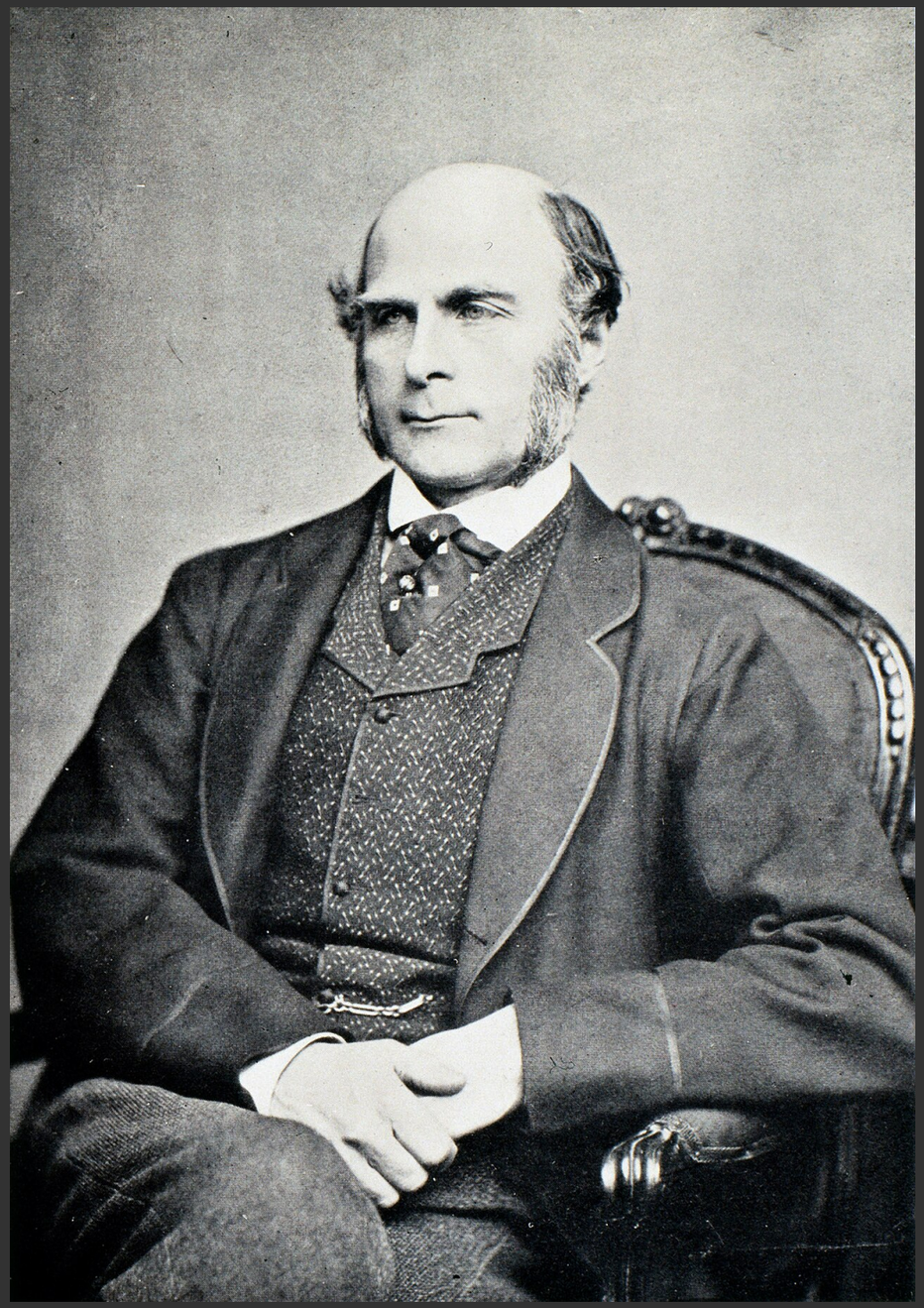
“we thus arrive in inquiries of this kind, at the fundamental principle: that the greater the number of individuals observed, the more do individual particularities, whether physical or moral, become effaced, and leave in a prominent point of view the general facts, by virtue of which society exists and is preserved.”

— Adolphe Quetelet, 1835

In the preceding quote, Quetelet lays down the philosophical grounding for what an average is and what claims we can make about that data. Statistically we can define a common or average as a typical specimen

# Sir Francis Galton (1822-1911)

- English Statistician, psychologist and anthropologist
- Cousin of Charles Darwin and sought to bring Darwin's theories into the study of society
- Coined the term eugenics



# A tale of two exhibitions

**Exhibit of American Negroes at the Paris World's Fair (1900)**

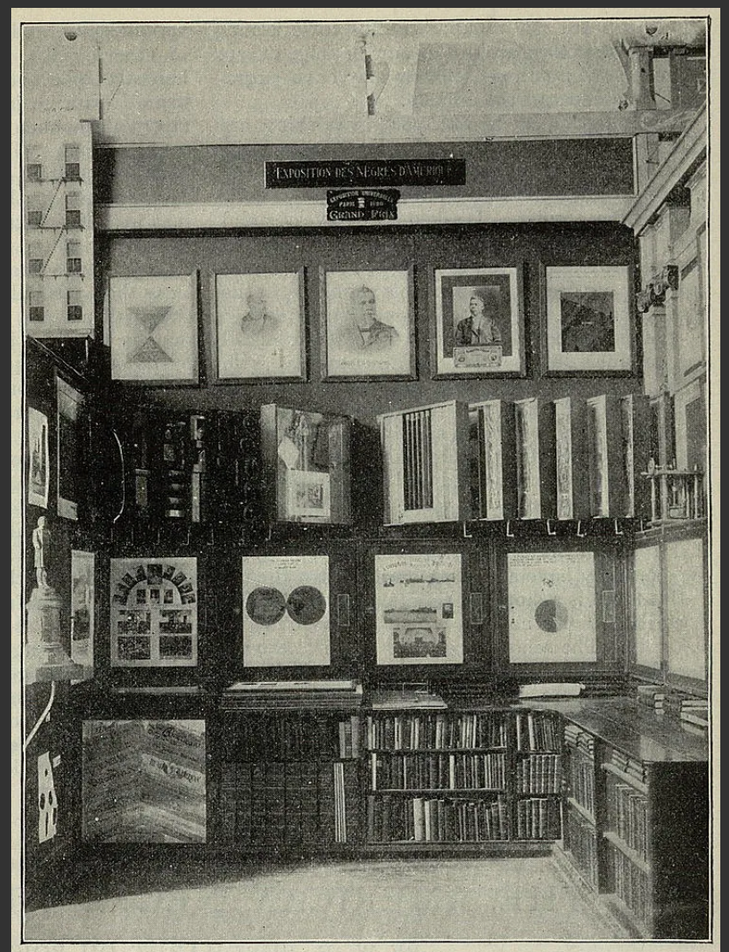


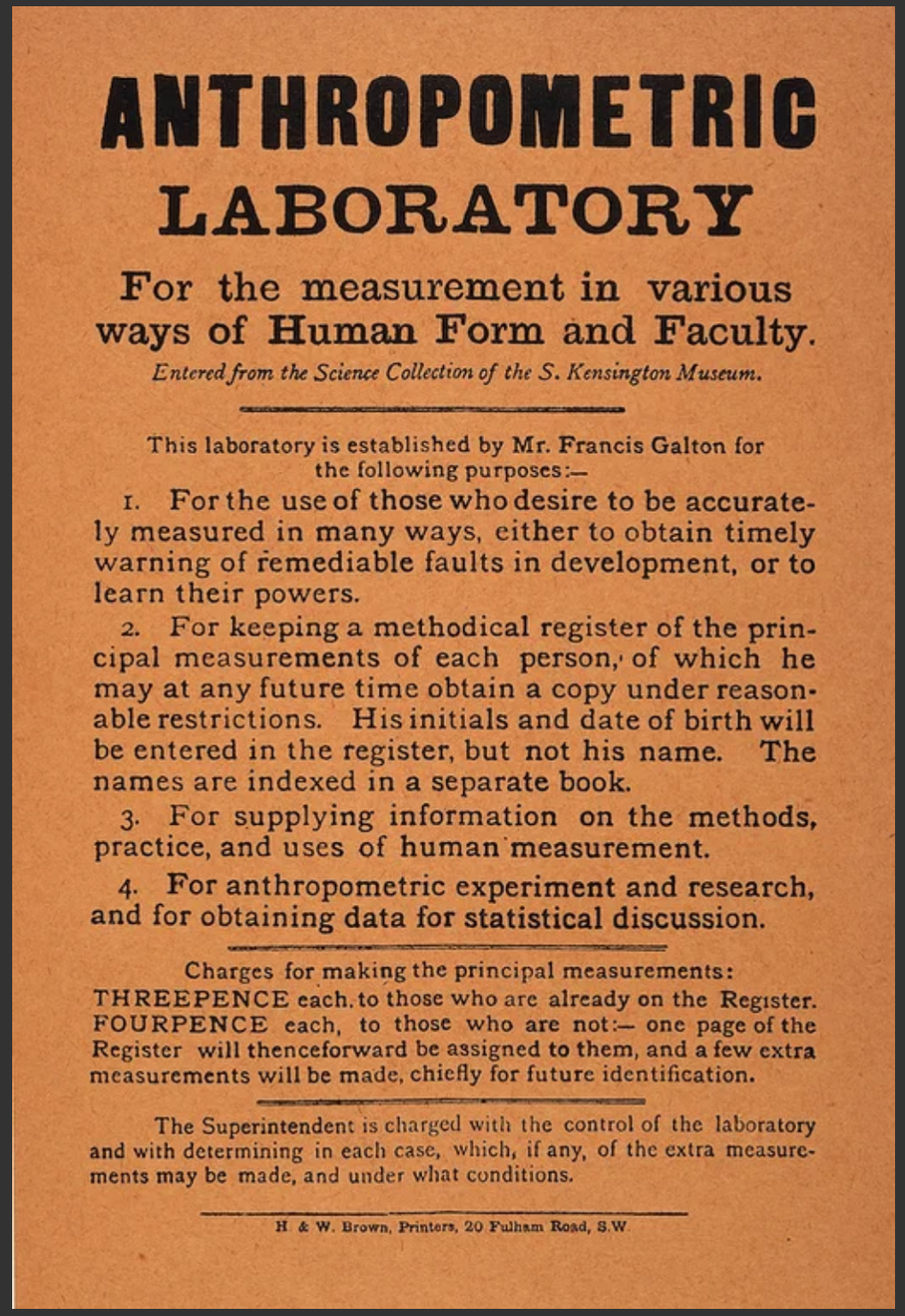

EXHIBIT OF AMERICAN NEGROES AT THE PARIS EXPOSITION.

**Galton’s Anthropometric Laboratory at the 1884 International Health Exhibition at South Kensington**

# The Anthropometric Laboratory

International Health Exhibition, South Kensington, 1884

- 9,337 people measured in 17 ways (Galton, 1885, p. 206)
- About 90 people a day in a space 36 feet long and 6 feet wide
- Visitors paid 3d. and received a card of their results
- Measured: sight, color sense, judgment of eye, hearing, breathing, strength, swiftness of blow, arm span, height, weight



ANTHROPOMETRIC
LABORATORY

For the measurement in various ways of Human Form and Faculty.

*Entered from the Science Collection of the S. Kensington Museum.*

This laboratory is established by Mr. Francis Galton for the following purposes:—

1. For the use of those who desire to be accurately measured in many ways, either to obtain timely warning of remediable faults in development, or to learn their powers.

2. For keeping a methodical register of the principal measurements of each person, of which he may at any future time obtain a copy under reasonable restrictions. His initials and date of birth will be entered in the register, but not his name. The names are indexed in a separate book.

3. For supplying information on the methods, practice, and uses of human measurement.

4. For anthropometric experiment and research, and for obtaining data for statistical discussion.

Charges for making the principal measurements:
THREEPENCE each, to those who are already on the Register.
FOURPENCE each, to those who are not:— one page of the Register will thenceforward be assigned to them, and a few extra measurements will be made, chiefly for future identification.

The Superintendent is charged with the control of the laboratory and with determining in each case, which, if any, of the extra measurements may be made, and under what conditions.

H. & W. Brown, Printers, 20 Fulham Road, S.W.

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4. For anthropometric experiment and research, and for obtaining data for statistical discussion.

**“The theory of hereditary genius, though usually scouted, has been advocated by a few writers in the past as well as in modern times. But I may claim to be the first to treat the subject in a statistical manner, to arrive at numerical results, and to introduce the “law of deviation from average” into discussions on heredity.”**

Galton, F. (1869). Hereditary genius: An inquiry into its laws and consequences. Macmillan.

# Two Ways of Measuring

| | Du Bois (Paris, 1900) | Galton (London, 1884) |
|---|---|---|
| Measures | Social conditions: education, land, occupation | Individual bodies and faculties: height, strength, senses |
| Presented as | Hand-drawn charts and photographs | Instruments, cards, a schedule |
| Explanation sought | Social and historical causes | Heredity (in his later work, e.g. *Hereditary Genius*) |

# Measurement as Mode of Discipline

“The use of periodical measurements is two-fold, personal and statistical. The one shows the progress of the individual; the other that of portions of the nation, or of the nation as a whole.” (p. 213)

Galton proposed periodic measurement as “a recognised part of school discipline” (p. 207)

# Pearson Correlation and Regression Toward the Mean

- Pearson correlation is widely used in statistics in the social sciences. Sir Francis Galton developed techniques for correlation and regression to study heredity in populations.
- Further developed by Karl Pearson to support his analysis of data and arguments for the intellectual superiority of the Aryan race.

# Mathematics of the Pearson Correlation

The Pearson correlation coefficient measures how strongly two variables move together:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

$r$ ranges from $-1$ to $1$. It has no units, so measurements in inches and pounds can be compared on one scale.

# Regression Toward the Mean

When one sample of a random variable is extreme then the next sample is likely to be closer to its mean. Galton argued that this illustrated a reversion to mediocrity in populations.

This can be understood in terms of intelligence: over the course of generations without intervention, intelligence in *good stock* will regress to the mean.

**Pearson correlation functions as an instrument for measuring sameness and difference. Originally, it was developed for explicitly racialized and eugenicist reasons.**

# Galton believed heredity could be analyzed probabilistically

- Regression (toward the mean) **the offspring tend toward the mean of the race**
- Correlation **one factor's dependency on another could be measured (even if they are in different, seemingly incommensurable units)**

# Karl Pearson (1857-1936)

- English Biostatistician and Mathematician, founded the world's first statistics department at University College London in 1911
- Developed many statistical techniques, including PCA, the chi-squared distribution, contingency tables, histograms, the mode, and the Pearson distribution
- Proponent of social darwinism and eugenics, mentee and biographer of Galton

Karl Pearson with Sir Francis Galton (1910)

# Principal Component Analysis and Dimensionality Reduction

PCA is a technique for dimensionality reduction that transforms data to a simpler representation that retains the most important information. Often used in exploratory data analysis.

Principal component analysis was first developed as a technique by Karl Pearson in 1901

# Principal Component Analysis

PCA finds patterns of variation across measured variables and summarizes them as new components.

But a component is not simply “in” the people being measured. It depends on:

- Which people are included
- Which traits are measured, and how
- Which variables are combined or standardized
- Which patterns the analysis treats as important

The mathematics works on the data it is given. It cannot tell us whether the data collection choices were fair or meaningful.

# How Choices Become Social Facts

Consider a study that measures bodies and searches for patterns among them.

Choosing the population, the measurements, and the units shapes the covariance matrix. PCA then returns components that are mathematically optimal for that matrix. If researchers interpret those components as innate differences between groups, the analysis can make those differences appear discovered rather than constructed through choices about measurement and interpretation.

# Biometrics, PCA, and Eugenic Reasoning

Karl Pearson developed his 1901 method for fitting lines and planes to observations within a broader biometric project of measuring variation in populations. Pearson was also a eugenicist.

# Alphonse Bertillon (1853-1914)

- French criminologist and biometrics researcher
- Developed and standardize the mug shot and created the first criminal photo database from anthropometric data

**“Every measurement slowly reveals the workings of the criminal. Careful observation and patience will reveal the truth.”**

– Alphonse Bertillon

## Francis Galton

individual $\rightarrow$ type

Galton sought to inductively discover types through statistical studies of heredity to discover how some led to the elevation and others to the decline of a race or society.

## Alphonse Bertillon

type $\rightarrow$ individual

Using photography, Bertillon developed a “database” of criminal types based on the measurement of physical attributes that could be used to convict individuals.

# Facial Recognition Technology, Bias, and the Politics of Recognition

# The Mathematics of Facial Recognition

From image to mathematical representation and computational object

### 1. Image as matrix

A photo is a grid of pixel values, $I \in \mathbb{R}^{h \times w \times 3}$

### 2. Face as vector

Software detects landmarks and extracts features, converting a face into a vector of numbers (an "embedding"): $f(I) \in \mathbb{R}^n$

### 3. Similarity as distance

Two faces are compared by the distance between their vectors:

$$d(u, v) = \|u - v\|$$

### 4. Identity as threshold

A match is declared when $d(u, v) < \tau$ for some threshold $\tau$

# The Geometry of the Face

A face can be treated as a set of points in space

**2D landmarks:** $p_i = (x_i, y_i)$ for eye corners, nose tip, mouth corners, jawline

**3D mesh:** vertices $V \in \mathbb{R}^{n \times 3}$ connected into triangles, approximating the surface of the face

**Geometric measures:** distances and ratios between points

**Alignment:** rotating, scaling, and translating the points to a standard pose before comparison

# Bertillon's Measurements, Automated

Bertillon recorded fixed bodily measurements (head length, ear, nose) and described faces with a standardized vocabulary. Facial recognition systems also turn a face into measured geometry, but compute it automatically and at massive scale.

# Apple Face ID: Depth and Mesh

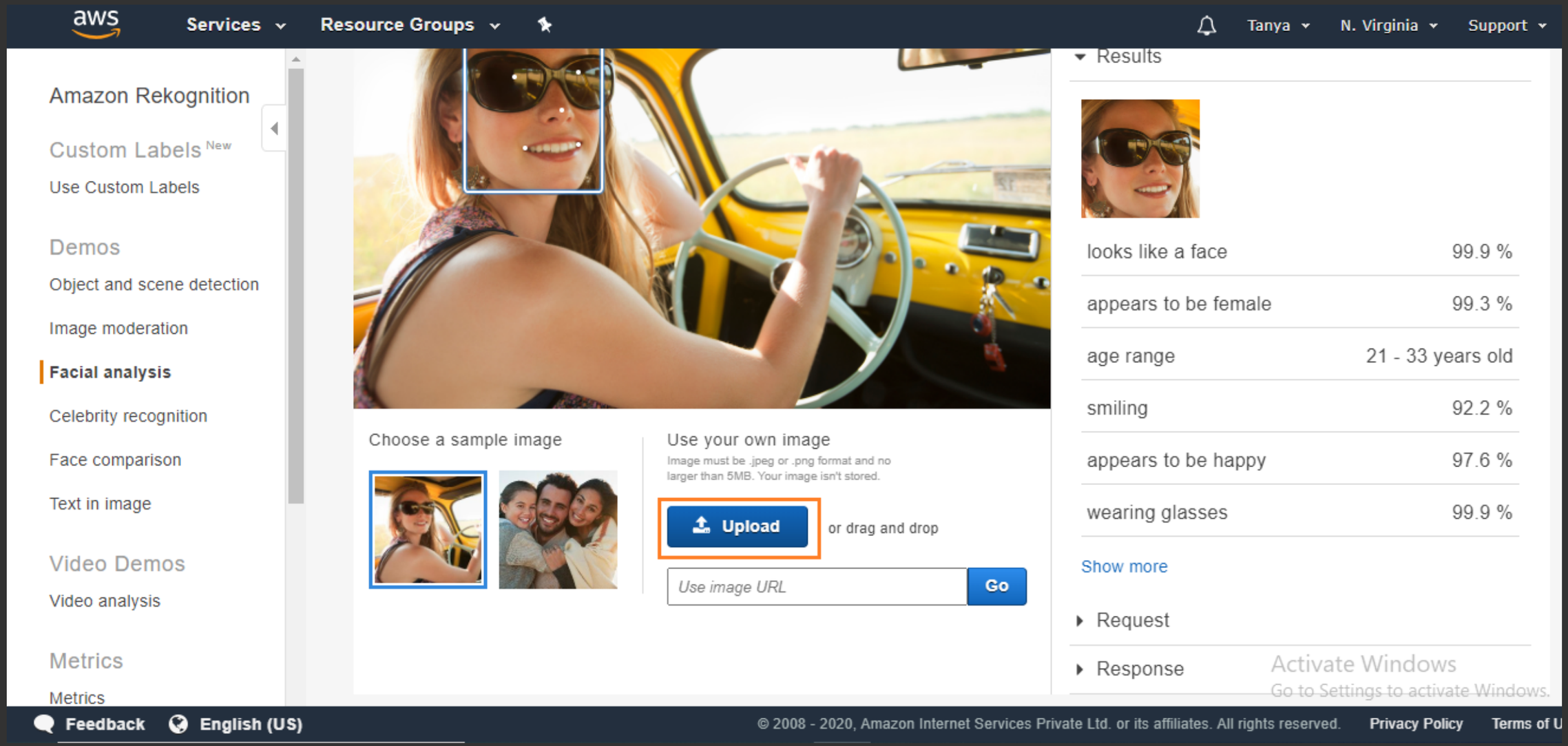
- **Projection:** the TrueDepth camera projects over 30,000 invisible infrared dots onto the face
- **Depth map:** distortion of the dot pattern gives a distance $z$ for each point, so the face becomes a surface $z = D(x, y)$
- **Mathematical representation:** a neural network converts the depth and infrared data into a numerical template, stored in the Secure Enclave on the device
- **Comparison:** a new scan is compared to the stored template, with adaptation over time

# Amazon Rekognition: 2D Landmarks

Rekognition works from ordinary 2D photographs, not depth sensors. Detect Faces returns:

- **Bounding box** around the face
- **Landmarks:** coordinates for eyes, nose, mouth, and face outline, given as ratios of image width and height
- **Pose:** roll, yaw, and pitch (the 3D rotation of the head estimated from a 2D image)
- **Quality:** brightness and sharpness
- **Attributes with confidence scores:** such as gender, age range, and emotion

# Amazon Rekognition



# Detect Faces API

Rekognition detects faces in images and videos and identifies facial landmarks and other attributes. Rekognition then returns confidence scores for each attribute

```
  "BoundingBox": {
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    "Height": 0.7510867118835449,
    "Left": 0.08881539851427078,
    "Top": 0.151064932346344
  },
  "AgeRange": {
    "Low": 18,
    "High": 26
  },
  "Smile": {
    "Value": false,
    "Confidence": 89.77348327636719
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  "Eyeglasses": {
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  "Sunglasses": {
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    "Confidence": 99.85968780517578
  },
  "Beard": {
```

# Dr. Joy Buolamwini

- Computer scientist, founder of the Algorithmic Justice League
- Face detection software failed to detect her face until she put on a white mask
- *Gender Shades* (Buolamwini & Gebru, 2018) tested gender classification from IBM, Microsoft, and Face++

# Gender Shades (2018)

- Common benchmark datasets skew male and lighter-skinned (Labeled Faces in the Wild: about 77% male, over 80% white).
- The Pilot Parliaments Benchmark uses parliamentarians from three African and three European countries, balanced by gender and skin type.
- Error rates: as high as 35% for darker-skinned women, no more than 1% for lighter-skinned men.

Proceedings of Machine Learning Research 81:1–15, 2018 Conference on Fairness, Accountability, and Transparency

## Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*

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**Editors:** Sorelle A. Friedler and Christo Wilson

### Abstract

Recent studies demonstrate that machine learning algorithms can discriminate based on classes like race and gender. In this work, we present an approach to evaluate bias present in automated facial analysis algorithms and datasets with respect to phenotypic subgroups. Using the dermatologist approved Fitzpatrick Skin Type classification system, we characterize the gender and skin type distribution of two facial analysis benchmarks, IJB-A and Adience. We find that these datasets are overwhelmingly composed of lighter-skinned subjects (79.6% for IJB-A and 86.2% for Adience) and introduce a new facial analysis dataset which is balanced by gender and skin type. We evaluate 3 commercial gender classification systems using our dataset and show that darker-skinned females are the most misclassified group (with error rates of up to 34.7%). The maximum error rate for lighter-skinned males is 0.8%. The substantial disparities in the accuracy of classifying darker females, lighter females, darker males, and lighter males in gender classification systems require urgent attention if commercial companies are to build genuinely fair, transparent and accountable facial analysis algorithms.

**Keywords:** Computer Vision, Algorithmic Audit, Gender Classification

### 1. Introduction

Artificial Intelligence (AI) is rapidly infiltrating every aspect of society. From helping determine who is hired, fired, granted a loan, or how long an individual spends in prison, decisions that have traditionally been performed by humans are rapidly made by algorithms (O'Neil, 2017; Citron and Pasquale, 2014). Even AI-based technologies that are not specifically trained to perform high-stakes tasks (such as determining how long someone spends in prison) can be used in a pipeline that performs such tasks. For example, while face recognition software by itself should not be trained to determine the fate of an individual in the criminal justice system, it is very likely that such software is used to identify suspects. Thus, an error in the output of a face recognition algorithm used as input for other tasks can have serious consequences. For example, someone could be wrongfully accused of a crime based on erroneous but confident misidentification of the perpetrator from security video footage analysis.

Many AI systems, e.g. face recognition tools, rely on machine learning algorithms that are trained with labeled data. It has recently been shown that algorithms trained with biased data have resulted in algorithmic discrimination (Bolukbasi et al., 2016; Caliskan et al., 2017). Bolukbasi et al. even showed that the popular word embedding space, Word2Vec, encodes societal gender biases. The authors used Word2Vec to train an analogy generator that fills in missing words in analogies. The analogy man is to computer programmer as woman is to "X" was completed with "homemaker", conforming to the stereotype that programming is associated with men and homemaking with women. The biases in Word2Vec are thus likely to be propagated throughout any system that uses this embedding.

* Download our gender and skin type balanced PPB dataset at gendershades.org

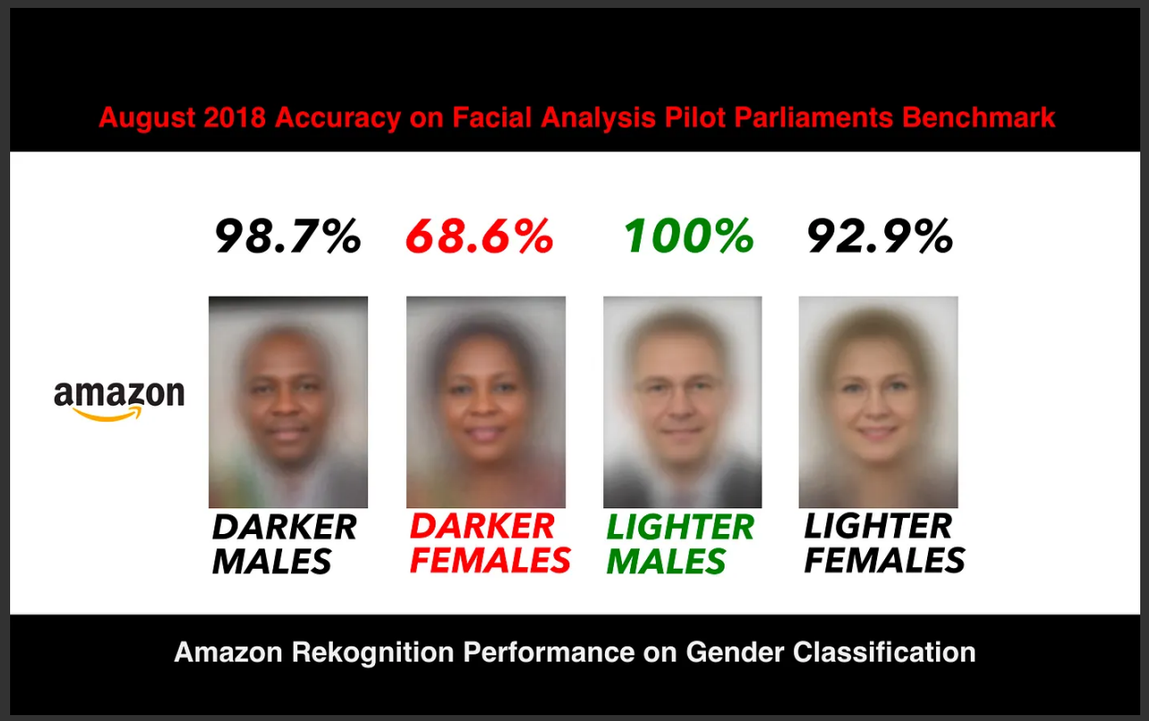
© 2018 J. Buolamwini & T. Gebru.

# Race and Gender Bias in Amazon Rekognition

Raji and Buolamwini (2019) audited Rekognition's gender classification with the same benchmark (data collected August 2018).

- Flawless performance on lighter-skinned men
- 68.6% accuracy on women of color (Buolamwini, 2019)

Amazon disputed the findings, saying the audit tested facial analysis, not facial recognition.



Amazon Rekognition Performance on Gender Classification

# Thresholds Are Choices

Rekognition returns a label with a confidence score. In Buolamwini's example, a photo of Oprah Winfrey was labeled "male" at 76.5%.

- Reported accuracy depends on the threshold chosen: IBM reported darker-skinned women going from 65.3% to 96.6% at a 0.99 threshold
- Amazon recommended 0.99 in one context and 0.95 in another; one police client reported not setting a threshold at all

# Where the Choices Are

Each step involves human decisions that get built into the system:

- **Training data:** whose faces were collected, and how they were labeled
- **Features:** which landmarks and attributes count
- **Threshold $\tau$:** how much error is acceptable, and for whom
- **Categories:** what the system is asked to classify (gender, age, emotion) and how these categories are constructed

**A mathematically precise output can hide the choices behind it, which is what Raji and Buolamwini's audit made visible.**

SECURITY AMAZON FACIAL RECOGNITION

# Amazon shareholders reject proposal to limit sales of AWS face recognition software

## 'Rekognition' ban voted down by 98% of shareholders

By James Miller May 25, 2019 at 9:52 AM | 7 comments | Add TechSpot

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**IN BRIEF:** Amazon has been at the center of the debate around facial recognition technology and its implementation by law enforcement.

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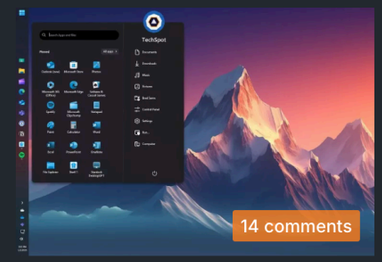


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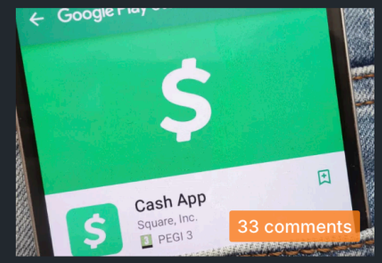


Woman spends three nights in jail and faces six years in prison over Cash App account she never opened

**Du Bois's work was not just a disagreement with the racialism of the social scientific practitioners of his day—it was an attempt to do the work of science differently.**

# Takeaways

- Measurement is a set of choices about what, who, how, and which categories, not a neutral recording
- Mathematical methods can make choices appear as if they are facts about nature
- Standard instruments, records, categories, and thresholds from in biometrics then show up in facial recognition now
- Du Bois shows data practice can be done differently

# Discussion

- What should we do about the racialized history of modern measurement and quantification methods?
- Who defines the categories being measured?
- What would it mean to do the work of measurement differently, as Du Bois did?

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